

PLSC 503: Homework 2

Extension to Acs (2025): Mapping the Political Contours of the Regulatory State:
Dynamic Estimates of Agency Ideal Points

Jody Park

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1. Replication Setup

Citations and Links

- **Original Paper:** ACS, A. (2025). Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points. *American Political Science Review*, 1-18.
- **Replication Data:** [R replication codes and dataset](#) from Acs (2025) “Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points (APSR).”
- **Paper for the Extension:** Fazekas, M., Ferrali, R., & Wachs, J. (2023). Agency Independence, Campaign Contributions, and Favoritism in US Federal Government Contracting. *Journal of Public Administration Research and Theory*, 33(2), 262-278.
- **Replication Data:** [R replication codes and dataset](#) from Fazekas et al. (2023) “Agency Independence, Campaign Contributions, and Favoritism in US Federal Government Contracting (JPART).”
- **Repository Links:**
 - [Dropbox folder](#); current code and data for this paper are in the *acs-2025-apsr* subfolder.
 - [GitHub repository for this project](#)

Article Introduction

Acs (2025) investigates the political ideology of US federal regulatory agencies by estimating dynamic ideal points for 69 administrative units across four presidential administrations (Clinton to Trump). The study uses data from the Unified Agenda on planned regulations and the president’s discretionary review of those regulations from the Office of Information and Regulatory Affairs (OIRA) docket. By analyzing the ideological positions of regulatory agencies over time, the paper provides insights into how agency ideology shapes regulatory behavior and how it interacts with political dynamics. The core contribution of the paper is to offer a comprehensive mapping of the political contours of the regulatory state, shedding light on the ideological underpinnings of regulatory activity and its implications for policy outcomes.

I extend the analysis by examining how agency ideology moderates the effects of political party contributions on favoritism in federal contracting, using the dynamic ideal point estimates from Acs (2025). Merging this agency ideology data with Fazekas et al. (2023) allows a direct test of whether ideological alignment between the president and an agency amplifies the returns to political contributions in procurement.

Data, Variables and Models

Data

In Acs (2025), the author uses publicly available datasets:

1. **Unified Agenda data:** Includes information on the regulations that federal agencies plan to implement, including their content, timing, and regulatory impact (a semi-annual catalog, indicating pending or completed).
2. **The Office of Information and Regulatory Affairs (OIRA) docket:** Lists the regulations reviewed. For each regulation, the docket includes information on the agency responsible, the date of review, and the outcome of the review (e.g., approved, disapproved, or returned for revision).
3. **Federal Register data:** Contains the official publication of federal regulations and related documents. [Federal Register API](#)

Acs (2025) uses the eight-digit regulation identifier number (the RIN) to merge the Unified Agenda data with the OIRA docket data.

- The analysis focuses on 69 administrative units (e.g., EPA, FDA, etc.) from 2001 to 2020, covering four presidential administrations (Clinton, Bush, Obama, Trump).
 - The author excludes regulatory agencies that “did not issue a regulation in each administration or issued fewer than 40 regulations during the entire period (averaging less than 10 per administration)” (Acs, 2025).
 - This exclusion criterion ensures that the analysis focuses on regulatory agencies with sufficient regulatory activity to estimate meaningful ideal points and examine their dynamics over time.

Variables

1. Dependent Variable:

- (1) **Review (Y_i)** (for the dynamic measurement model): Binary indicator for whether a planned rule in the Unified Agenda was reviewed by OIRA (1 = reviewed at least once in the OIRA docket; 0 = not reviewed).
- (2) **Regulatory Activity_{at}** (for regulatory productivity model): Agency-level regulatory output in administration t , measured using Federal Register final-rule page length (average per year within administration, with alternative coding rules reported in Table 2 and Supplementary Appendix).

2. Independent Variable:

- **President-Agency ideological distance:** $|X_p - X_{\{at\}}|$, where X_p is the president’s fixed ideal point (DW-NOMINATE anchors) and $X_{\{at\}}$ is the agency ideal point in administration t .

3. Model Components:

- **Rule significance (S_i):** Binary indicator from Unified Agenda priority categories (1 = economically significant/significant; 0 = otherwise), plus interaction with spatial distance in the measurement model.
- **Model structure:** **Agency intercepts (a_a)** and **dynamic structure:** Agency-specific baseline review propensity and a random-walk evolution of agency ideal points across administrations (Clinton, Bush, Obama, Trump), estimated in Stan (Supplementary Appendix B).

Models

For this homework, I will focus on *Model 2* (Regulatory Productivity) and *H3*, which use a negative binomial model to examine the relationship between President-Agency ideological distance and regulatory output.

- **Model 1 (Dynamic Measurement Model / IRT-Probit):** The core model estimates agency ideal points from OIRA review decisions using a Bayesian probit IRT structure with president-agency distance, significance terms, agency intercepts, and administration-specific agency ideal points that evolve via a random walk. The Stan specification and priors are provided in Supplementary Appendix B.
- **Model 2 (Validation Model: Regulatory Productivity):** The paper then estimates a *negative binomial model* (Equation 3 in the original paper) where regulatory activity is regressed on President-Agency Distance with agency fixed effects and multiple outcome constructions (Table 2), testing whether ideologically closer agencies are more productive.

$$\begin{aligned} \text{Regulatory Activity}_{at} &\sim \text{NegBin}(\mu_{at}, \theta) \\ \ln(\mu_{at}) &= \pi_a + \beta \cdot \text{PresidentAgencyDistance}_{at} \end{aligned} \tag{3}$$

Tested Hypotheses in the Original Paper:

- **H1 (Review-distance expectation):** OIRA review probability increases as ideological distance between the president and an agency increases.
- **H2 (Dynamic ideology expectation):** Agency ideology is not static; agency ideal points shift across presidential administrations.
- **H3 (Productivity-distance expectation):** Regulatory productivity is lower when President-Agency ideological distance is larger (negative association in Equation 3 / Table 2).

2. Replication Accuracy

Figure 4. Conservative Trends in Agency Ideal Points

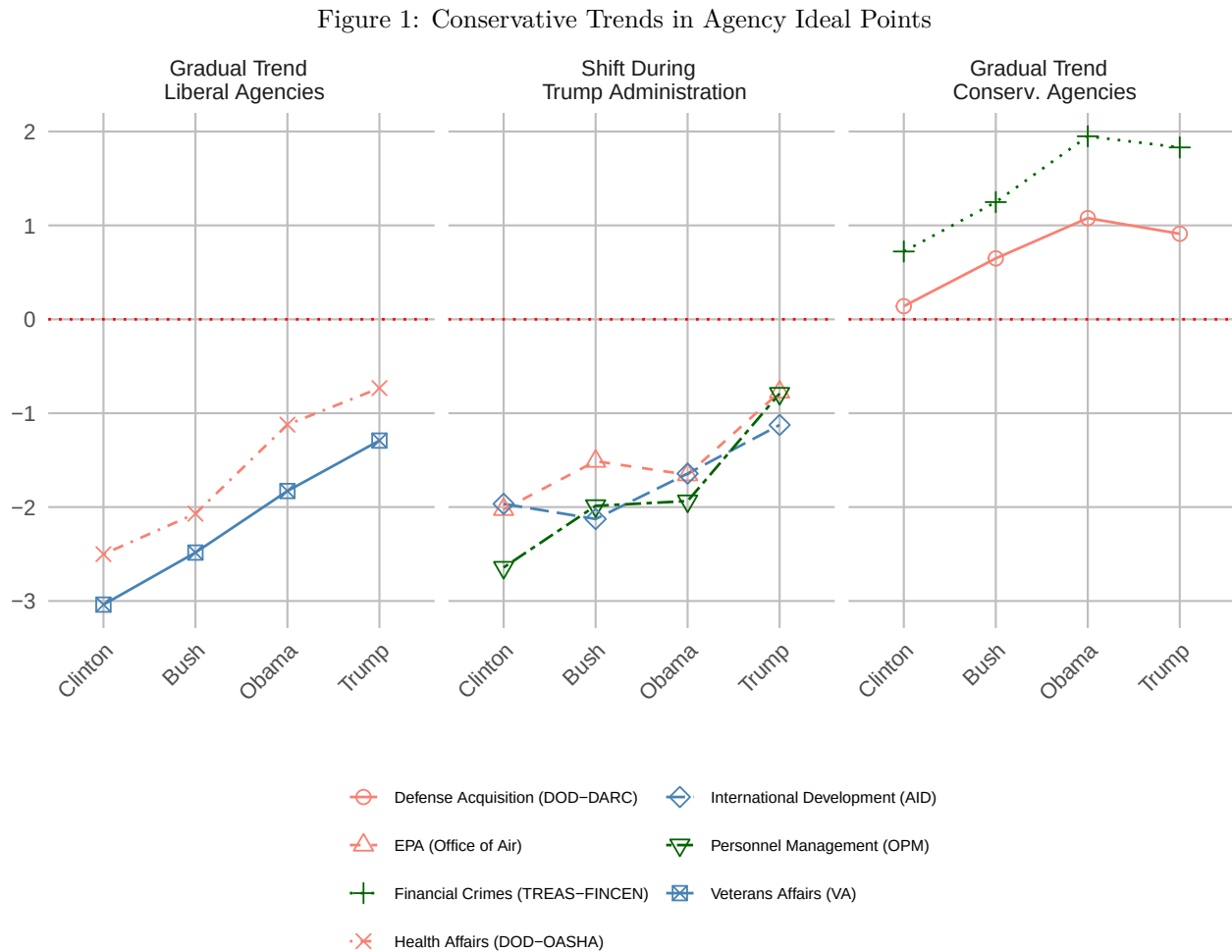


Figure 1 replicates the original pattern that several agencies shift in a conservative direction over administrations, with the largest movement concentrated in a small subset of agencies. This visual replication is consistent with Acs (2025)'s Figure 4.

Table 1. Agencies with Largest Shifts in the Conservative Direction

Table 1: Agencies with Largest Shifts in the Conservative Direction

Agency	Ideal Points		Plausible Driver of Change
	Min	Max	
Personnel Management (OPM)	-2.64	-0.79	<i>Politicization:</i> Trump administration
EPA’s Office of Air (EPA-AIR)	-2.02	-0.78	Trump administration
Agency for Intl. Development (USAID)	-2.13	-1.13	Trump administration
Veterans Affairs (VA)	-3.04	-1.29	<i>New policy conflict:</i> VA MISSION Act of 2018, wars
Defense Acquisition (DOD-DARC)	0.14	1.08	Iraq/Afghan wars
Financial Crimes (TREAS-FINCEN)	0.72	1.95	Patriot Act of 2001, civil liberties
Health Affairs, Def. (DOD-OASHA)	-2.50	-0.73	Iraq/Afghan wars

Table 1 reproduces the agency-level minima and maxima from the Figure 4 subset and highlights the same conservative movers. The ordering and substantive interpretation follow the original table.

Table 2. Ideological Distance and Regulatory Productivity (Negative Binomial)

Table 2: Ideological Distance and Regulatory Productivity (Negative Binomial)

	Dependent variable: Regulatory activity (final rule page length)				
	(1)	(2)	(3)	(4)	(5)
	Formal	Formal significant	Formal significant regulatory	Formal (excl. OIRA)	Formal regulatory (excl. OIRA)
Pres.-Agency Distance	-0.048 (0.036)	-0.186* (0.049)	-0.218* (0.054)	-0.152* (0.036)	-0.218* (0.038)
Agency Fixed Effects	✓	✓	✓	✓	✓
Observations	276	276	276	276	276

* $p < 0.05$. Standard errors in parentheses. Models 4-5 exclude rules reviewed by OIRA at the proposed rule stage. Models 3 and 5 also exclude deregulatory rules. Weights equal to the inverse variance of each agency ideal point are included.

Table 2 replicates the negative-binomial validation result: larger President-Agency ideological distance is associated with lower regulatory productivity across outcome definitions. The sign and substantive magnitude patterns are aligned with the original paper.

Table A1. Agency Names and Ideal Point Estimates

Table 3: Agency Names and Ideal Point Estimates

agency id	clinton sig mean	clinton review mean	clinton n	bush43 sig mean	bush43 review mean	bush43 n	obama sig mean	obama review mean	obama n	trump sig mean	trump review mean	trump n
AG-AMS	0.229	0.200	35	0.125	0.125	64	0.286	0.167	42	0.222	0.222	18
AG-APHIS	0.826	0.783	46	0.676	0.648	71	0.394	0.394	66	0.150	0.150	20
AG-FNS	0.586	0.586	58	0.857	0.821	28	0.390	0.439	41	0.692	0.769	13
AG-FS	0.217	0.304	23	0.500	0.412	34	0.375	0.375	16	0.625	0.500	8
AG-FSIS	0.543	0.478	46	0.375	0.250	24	0.458	0.458	24	0.176	0.176	17
AG-GIPSA	0.348	0.348	23	0.059	0.059	17	0.364	0.364	11	1.000	1.000	1
AG-RHS	0.450	0.450	20	0.154	0.154	13	0.167	0.083	12	0.000	0.000	9
AG-RUS	0.194	0.226	31	0.174	0.130	23	0.083	0.083	12	0.000	0.000	1
AID	0.167	0.333	6	0.667	0.889	9	0.625	0.500	16	0.444	0.444	9
COM-BEA	0.000	0.000	13	0.000	0.080	25	0.000	0.000	12	0.000	0.200	5
COM-BIS	0.800	0.800	5	0.400	0.300	20	0.721	0.744	43	0.750	0.750	8
COM-ITA	0.143	0.143	7	0.200	0.200	15	0.118	0.059	17	1.000	1.000	4
COM-NOAA	0.060	0.101	398	0.125	0.133	534	0.061	0.051	528	0.067	0.067	252
COM-PTO	0.050	0.150	20	0.289	0.316	38	0.654	0.692	26	0.214	0.357	14
DHS-USCBP	0.667	0.000	3	0.565	0.391	23	0.357	0.357	14	0.273	0.182	11
DHS-USCG	0.267	0.000	15	0.315	0.185	54	0.143	0.125	56	0.111	0.111	18
DHS-USCIS	1.000	0.000	8	1.000	0.700	30	1.000	1.000	10	0.889	0.722	18
DOJ-DARC	0.022	0.022	90	0.050	0.050	141	0.280	0.318	132	0.202	0.112	89
DOJ-DOAOASHA	0.036	0.393	28	0.484	0.839	31	0.212	0.182	33	0.125	0.125	8
DOE-EE	0.444	0.250	36	0.406	0.375	32	0.361	0.340	147	0.316	0.368	38
DOI-BIA	0.239	0.109	46	0.136	0.364	22	0.231	0.231	13	0.000	0.000	7
DOI-BLM	0.206	0.206	34	0.235	0.235	17	0.462	0.615	13	0.500	0.625	8
DOI-FWS	0.077	0.181	310	0.406	0.206	310	0.091	0.103	331	0.100	0.100	120
DOI-MMS	0.157	0.255	51	0.178	0.178	45	0.545	0.273	11	1.000	1.000	1
DOI-NPS	0.222	0.361	36	0.286	0.306	49	0.139	0.139	36	0.188	0.250	16
DOJ-BOP	0.125	0.031	32	0.750	0.700	20	0.533	0.467	15	0.333	0.333	3
DOJ-DEA	0.250	0.333	24	0.710	0.871	31	0.727	0.727	11	0.800	0.600	10
DOJ-EOIR	0.222	0.111	9	0.400	0.800	15	0.600	0.600	5	1.000	0.727	11
DOJ-LA	0.400	0.450	20	0.850	0.600	20	0.500	0.417	12	0.800	0.800	5
DOL-EBSA	0.588	0.647	17	0.850	0.900	20	0.964	0.857	28	1.000	0.833	12
DOL-ETA	1.000	1.000	9	1.000	0.889	18	1.000	0.917	12	1.000	1.000	8
DOL-MSHA	0.583	0.417	12	0.947	0.842	19	1.000	1.000	8	0.667	0.667	3
DOL-OSHA	0.773	0.273	22	0.722	0.833	18	0.435	0.304	23	0.889	0.889	9
DOL-PBGC	0.312	0.188	16	0.714	0.643	14	0.308	0.385	13	0.000	0.000	4
DOT-FAA	0.422	0.296	135	0.308	0.240	104	0.267	0.280	75	0.304	0.261	23
DOT-FHWA	0.308	0.231	78	0.289	0.289	38	0.419	0.387	31	0.417	0.250	12
DOT-FMCSA	0.611	0.139	36	0.632	0.632	19	0.464	0.393	28	0.200	0.240	25
DOT-FRA	0.652	0.478	23	0.348	0.261	23	0.342	0.316	38	0.250	0.250	16
DOT-FTA	0.389	0.167	18	0.375	0.188	16	0.500	0.450	20	0.333	0.000	3
DOT-NHTSA	0.107	0.114	140	0.228	0.228	101	0.304	0.261	69	0.267	0.333	15
DOT-OST	0.255	0.218	55	0.293	0.317	41	0.375	0.417	24	0.417	0.417	12
DOT-PRMISA	0.222	0.159	63	0.280	0.240	50	0.327	0.269	52	0.500	0.500	10
ED-OESE	0.800	0.600	5	0.889	0.889	9	0.727	0.818	22	0.000	0.000	4
ED-OPE	0.679	0.464	28	1.000	1.000	5	0.955	0.955	22	1.000	0.818	11
EPA-AR	0.361	0.354	291	0.374	0.505	281	0.525	0.506	263	0.382	0.382	123
EPA-OPPTS	0.591	0.455	44	0.333	0.381	21	0.558	0.538	52	0.500	0.607	28
EPA-SWER	0.650	0.567	60	0.771	0.686	35	0.484	0.484	31	0.882	0.765	17
EPA-WATER	0.412	0.425	80	0.362	0.468	47	0.488	0.488	41	0.706	0.647	17
FAR	0.171	0.238	181	0.124	0.114	105	0.621	0.589	95	0.317	0.366	41
GSA	0.323	0.355	62	0.125	0.083	48	0.171	0.122	41	0.667	0.533	15
HHS-ACF	0.548	0.714	42	0.833	1.000	18	0.929	0.857	14	0.857	1.000	7
HHS-CMS	0.718	0.825	103	0.897	0.960	174	0.985	0.923	196	0.978	0.944	90
HHS-FDA	0.620	0.479	121	0.726	0.500	62	0.618	0.500	68	0.500	0.525	40
HUD-OH	0.548	0.595	42	0.821	0.821	56	0.440	0.460	50	0.625	0.625	8
HUD-PHI	0.949	0.846	39	0.867	0.833	30	0.706	0.588	17	1.000	1.000	5
NARA	0.389	0.389	18	0.194	0.258	31	0.625	0.562	16	0.333	0.333	3
NASA	0.111	0.111	9	0.059	0.059	17	0.364	0.182	33	0.333	0.333	6
NCUA	0.000	0.000	31	0.014	0.000	72	0.046	0.000	65	0.000	0.000	38
OPM	0.136	0.544	103	0.558	0.865	104	0.528	0.613	106	0.684	0.447	38
RRB	0.097	0.097	31	0.385	0.385	13	0.200	0.000	5	0.000	0.000	2
SBA	0.627	0.520	75	0.784	0.745	51	0.614	0.649	57	0.385	0.269	26
SSA	0.439	0.316	57	0.775	0.800	80	0.895	0.684	38	0.800	0.700	10
STATE	0.111	0.333	9	0.405	0.514	37	0.776	0.690	58	0.857	0.857	21
TREAS-CUSTOMS	0.015	0.000	66	0.000	0.000	22	0.278	0.222	18	0.714	0.571	7
TREAS-DO	0.167	0.000	12	0.275	0.250	40	0.571	0.571	21	0.400	0.100	10
TREAS-FINCEN	0.000	0.000	11	0.062	0.031	32	0.630	0.593	27	0.750	0.250	4
TREAS-OCC	0.023	0.047	43	0.278	0.083	36	0.256	0.070	43	0.179	0.000	39
TREAS-TTB	0.000	0.000	7	0.055	0.018	55	0.050	0.050	20	0.500	0.500	2
VA	0.228	0.596	114	0.580	0.902	112	0.409	0.299	127	0.343	0.388	67

Table 3 reproduces the appendix-style agency-by-administration summaries of significance rates, review rates, and counts. These summaries support the data pipeline used to estimate the main replication models.

3. Methodological Extension

Why this method is necessary

The ACS validation model uses regulatory activity (final-rule page length) as the dependent variable to measure agency productivity.

For this homework, I shift the dependent variable to agency outcomes from Fazekas et al. (2023), specifically donation and contracting outcomes, while retaining the ACS ideological distance measure as the main explanatory variable. This directly tests whether president-agency ideological alignment is associated with *who receives contracts* and *how politically active those contractors are*, not only with how many pages of rules are produced as a measure of agency productivity.

How the method improves the paper

This extension keeps the ACS measure of president-agency ideological distance but changes the outcome from rule production to procurement outcomes. In the main extension models, I also add ACS-side controls for rule volume, significant rules, and review rates (`n_rules`, `n_sig_rules`, `review_rate`) so the estimated relationship is less likely to reflect simple differences in regulatory workload.

Extension 1: Logistic Regression on Donation Probability

Acs (2025) establishes that ideological distance between the president and an agency depresses regulatory output. A parallel mechanism operates on the firm side: if firms anticipate that ideologically close agencies are more responsive to political pressure, they donate when distance is small. I test this directly by regressing the binary donation indicator `isDonation` on `distance` using logistic regression with agency fixed effects, following Chapter 13 of Gelman et al. (2025):

$$\Pr(\text{isDonation}_{ift} = 1) = \text{logit}^{-1}(\alpha + \beta \cdot \text{distance}_{at} + \pi_a)$$

where π_a is an agency fixed effect and `distanceat` is the ACS absolute president-agency ideological distance. A negative $\hat{\beta}$ indicates that firms donate more when the agency's ideology is close to the president, consistent with strategic targeting of politically aligned agencies. This model requires a binomial distribution and logit link; the divide-by-4 rule translates the logit coefficient into an approximate marginal probability effect. The binary nature of `isDonation` makes the negative binomial distribution from ACS Equation (3) inappropriate, so logistic regression is the correct GLM for this outcome.

Extension 2: Poisson versus Negative Binomial Overdispersion Diagnostic

Before extending the main results, I validate the negative binomial specification by reestimating ACS Equation (3) with a Poisson model and comparing fit using posterior predictive checks (Gelman et al., 2025). The Poisson distribution is the nested special case of the negative binomial as the reciprocal dispersion parameter $\phi \rightarrow \infty$.

For count outcomes, the Poisson model imposes $\text{sd}(y_i) = \sqrt{\text{E}(y_i)}$, which is violated whenever the data show more variance than this constraint allows. I generate 200 simulated replications y^{rep} from both fitted models using `simulate()` on frequentist Poisson and negative binomial models fit with `glm` and `MASS::glm.nb`, and compare the frequency of zero counts in replicated data to the observed distribution.

A Poisson model, which systematically underestimates zeros with replicated zero-rates far below the observed proportion, confirms that overdispersion is real and that the negative binomial specification in Acs (2025) is necessary. This diagnostic is independent of the main extension and applies directly to the replication sample.

Extension 3: Ordered Logistic Regression on Agency Independence Tier

Acs (2025) estimates ideal points for agencies that span the full range of organizational independence. A remaining question is whether ideological distance predicts where an agency falls in Fazekas et al. (2023)’s four-tier independence hierarchy. I test this using ordered logistic regression, treating `agency_cat2` (1 = most independent, 4 = least independent) as an ordered categorical outcome:

$$\Pr(\text{agency_cat}_a > k) = \text{logit}^{-1}(\beta \cdot \text{distance}_a - c_k), \quad k = 1, 2, 3$$

where c_k are cutpoints estimated alongside β .

Fazekas et al. (2023)’s four-tier agency independence hierarchy:

- (1) `indep. commission & IRC`: most independent agencies
- (2) `indep. agency`: independent but not commission-based
- (3) `caninet/exec. dep. and bureau`: agencies with some independence but also more direct presidential oversight
- (4) `caninet/exec. dep., not bureau`: least independent agencies

A negative $\hat{\beta}$ implies that agencies with greater ideological distance from the president tend to occupy more independent tiers, consistent with the institutional logic that independent agencies face less direct presidential oversight. Here, the levels go from 1 (most independent) to 4 (least independent).

I fit this using `MASS::polr`, which estimates β and the cutpoints c_k by maximum likelihood. This model is appropriate because the four tiers are ordered (independence is a latent continuous construct bucketed into discrete categories) but the gaps between tiers are not necessarily equal, which rules out treating `agency_cat` as a continuous dependent variable in OLS.

Explanatory Tables and Figures

Table 4: ACS distance + Fazekas outcomes

Variable	Role	Definition
<code>distance</code>	Independent variable (ACS)	Absolute president-agency ideological distance from ACS dynamic estimates
<code>donor_contract_share</code>	Dependent variable (Fazekas)	Share of contracts in agency-admin cell awarded to donor firms
<code>donor_contract_count</code>	Dependent variable (Fazekas)	Count of donor contracts in agency-admin cell
<code>donation_amount</code>	Dependent variable (Fazekas)	Total donation amount linked to firms in agency-admin cell
<code>n_contracts</code>	Dependent variable (Fazekas)	Total number of contracts in agency-admin cell
<code>n_rules</code>	Control / alternative productivity measure (ACS)	Number of planned rules in ACS estimation data
<code>n_sig_rules</code>	Control / alternative productivity measure (ACS)	Number of significant planned rules in ACS estimation data
<code>review_rate</code>	Control / alternative productivity measure (ACS)	Mean OIRA review indicator in ACS estimation data

Table 4 defines the merged-variable design used below. The dependent variables are now donation and contracting outcomes from Fazekas et al. (2023), while ACS ideological distance remains the core predictor.

Table 5: Merge diagnostics for ACS-Fazekas extension panel

Metric	Value
Agency-admin rows in Fazekas aggregate (Bush/Obama)	317
Rows matched to ACS distance	48
Distinct matched agencies	25
Mean donor contract share	0.138
Mean contracts per matched agency-admin row	861.208

Table 5 reports the merge coverage for the new agency-administration panel. The matched panel is the estimation sample used in the extension models below.

Table 6: ACS distance and Fazekas donation/contract outcomes

	Dependent variables from Fazekas (2023)			
	Donor contract share	Donor contract count	Log donation amount	Total contracts
President-Agency Distance	0.000 (0.019)	0.168 (0.137)	-2.256* (0.789)	-0.658* (0.333)
Obama administration (ref: Bush)	-0.045 (0.023)	-0.311* (0.143)	-0.242 (0.937)	0.383 (0.396)
Num.Obs.	48	48	48	48

* $p < 0.05$

Standard errors in parentheses. Unit of analysis is the agency-administration cell.

Table 6 extends ACS Equation (3) by replacing regulatory activity outcomes with donation and contracting outcomes from Fazekas et al. (2023). Columns 1 and 3 use `lm` for continuous outcomes, while columns 2 and 4 use `glm.nb` for count outcomes. A negative coefficient on `distance` implies that greater ideological distance is associated with lower donor-linked contracting intensity (or lower donation concentration), conditional on rule-volume controls.

Figure 2: Distance effect across alternative Fazekas outcomes

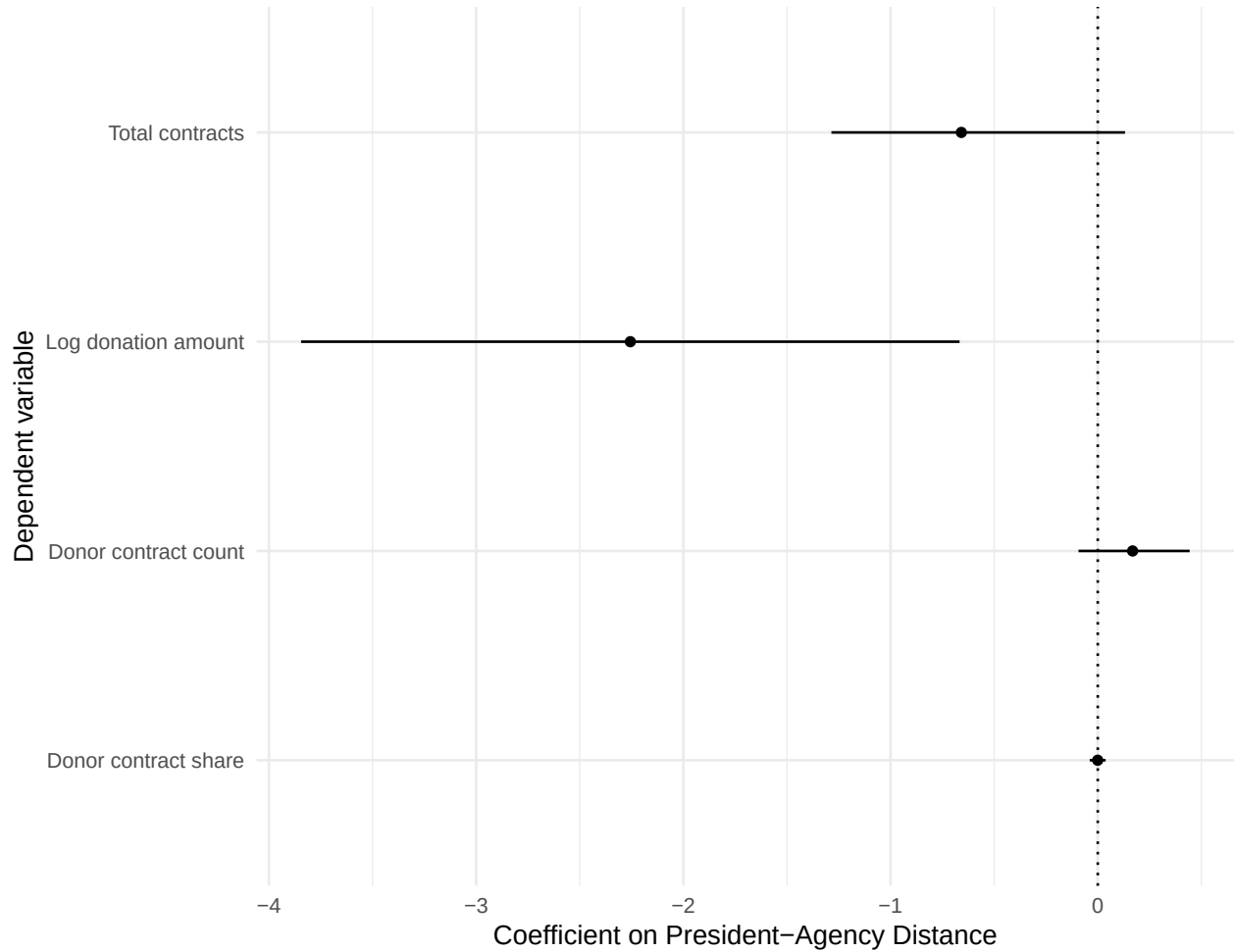


Figure 2 compares the `distance` coefficient across multiple dependent-variable definitions. This directly addresses sensitivity to outcome choice, which is one of the main robustness goals for this homework.

Extension 1: Logistic Regression on Donation Probability

Table 7: Logistic Regression of Donation Probability on Ideological Distance

	(1) No controls	(2) Admin FE
President-Agency Distance	0.097* (0.035)	0.191* (0.036)
Obama (ref: Bush)		-0.290* (0.032)
Observations	41 338	41 338
Log Likelihood	-14 792.631	-14 752.032

* $p < 0.05$

Standard errors in parentheses. Unit of analysis is the contract.

Table 7 reports logistic regression coefficients for `donation_dummy` on `distance`. A negative coefficient on `distance` indicates that firms are less likely to donate when ideological distance between the president and the agency is larger, consistent with strategic targeting of politically aligned agencies.

Figure 3: Predicted Probability of Donation by President-Agency Ideological Distance

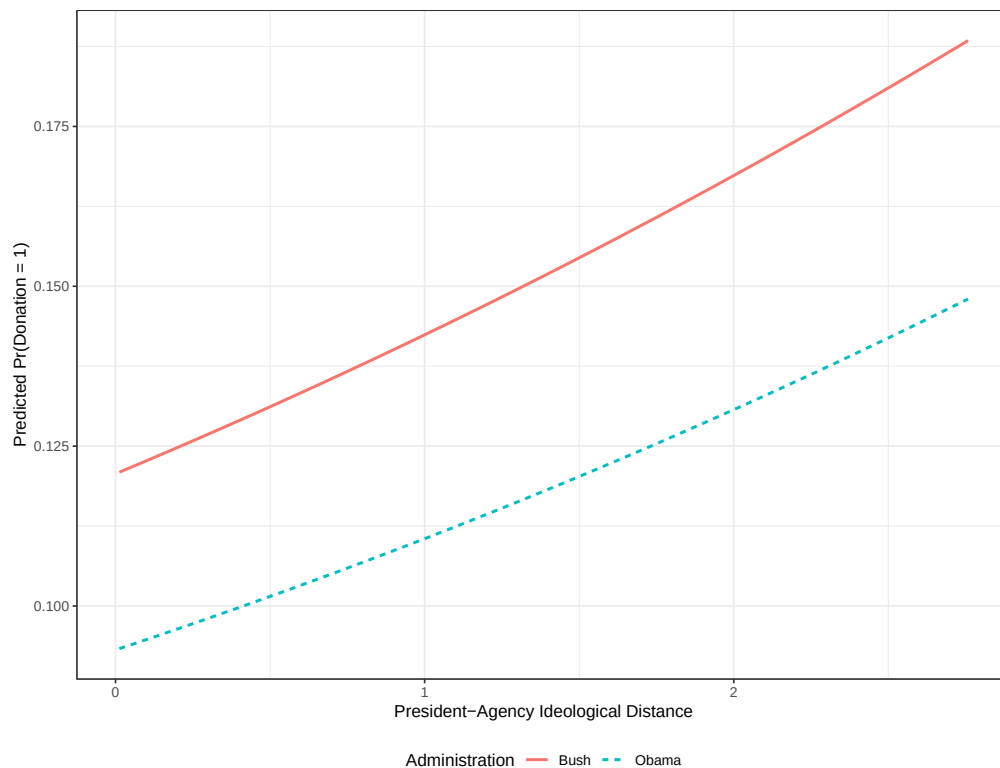


Figure 3 plots the predicted probability of donation as a function of `distance` for Bush and Obama administrations. The downward slope indicates that firms are less likely to donate as ideological distance increases.

Extension 2: Overdispersion Diagnostic

Figure 4: Posterior Predictive Check: Proportion of Zeros in Poisson vs. Negative Binomial Replications

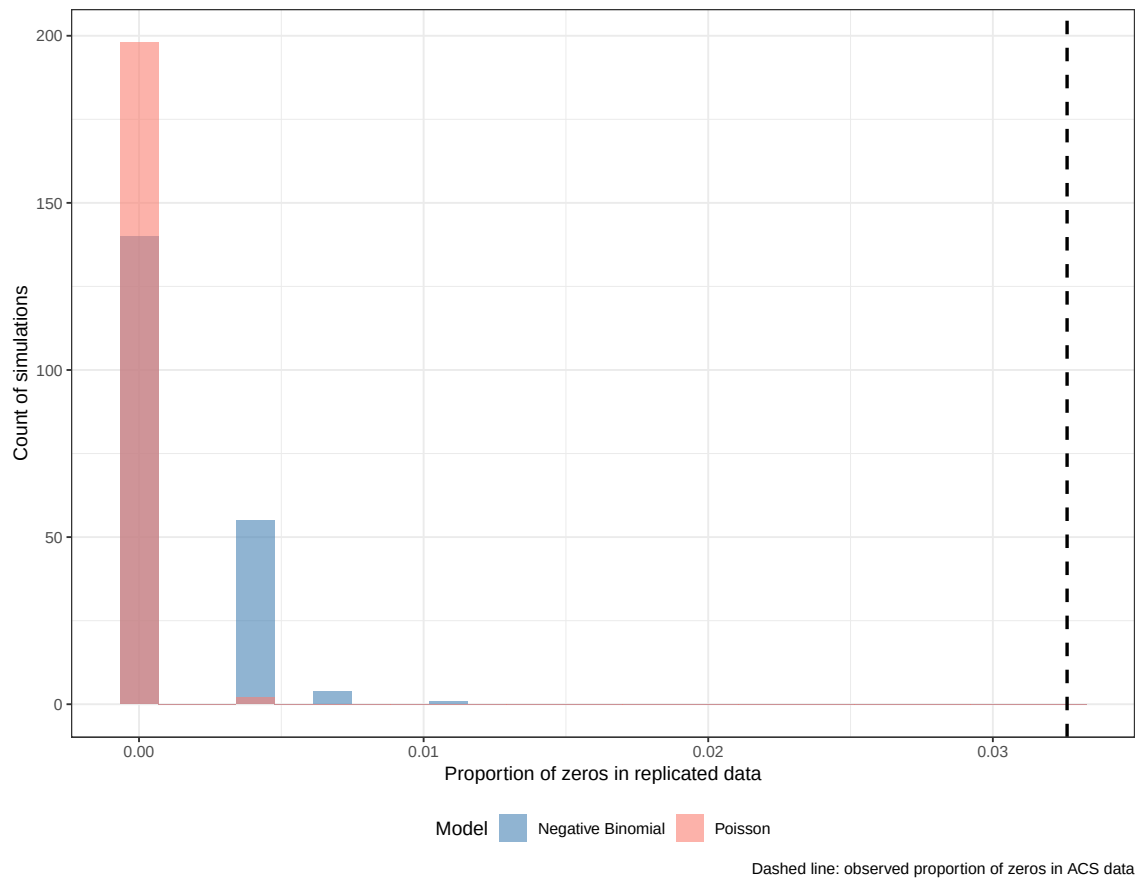


Figure 4 compares the proportion of zeros in 200 simulated datasets from Poisson and negative binomial models to the observed zero-rate (dashed line). The Poisson model systematically underestimates the proportion of zeros, while the negative binomial model generates replications closer to the observed rate, validating the negative binomial specification in Acs (2025).

Extension 3: Ordered Logistic Regression on Agency Independence Tier

Table 8: Ordered Logistic Regression of Agency Independence Tier on Ideological Distance

	Agency Independence Tier
President-Agency Distance	−0.665 (1.164)
Observations	42

* $p < 0.05$

Standard errors in parentheses. Outcome ordered from most independent to least independent. Cutpoints omitted.

Table 8 reports the ordered logistic coefficient for **distance** on the four-tier agency independence scale, from most independent (level 1: indep. commission and IRC) to least independent (level 4: cabinet department). A negative $\hat{\beta}$ implies that agencies with greater ideological distance from the president tend to fall into more independent (lower-numbered) tiers, consistent with the institutional logic that structurally independent agencies face less direct presidential alignment pressure. A null result would indicate that ideological distance and formal independence are orthogonal properties in this sample.

Extension 3a: Agency-Level Alternative

Extension 3a re-estimates Extension 3 after collapsing the panel to one row per agency, replacing administration-specific **distance** with each agency’s mean ideological distance across administrations. The motivation is statistical: agency independence tier is a fixed agency-level characteristic that does not vary across administrations, so treating each agency-administration row as an independent observation inflates the effective sample size in Extension 3. The agency-level specification corrects this by using the natural unit of the outcome variable. If the coefficient on mean distance is negative and close in magnitude to Extension 3, the cross-administration result is more credible as a stable cross-agency pattern rather than an artifact of repeated observations.

Table 9: Agency-Level Ordered Logistic Regression of Independence Tier on Mean Ideological Distance

	Agency Independence Tier
Mean President-Agency Distance	−1.324 (2.611)
Observations	22

* $p < 0.05$

Standard errors in parentheses. Outcome ordered from most independent to least independent. One row per agency.

Table 9 reports the same ordered-logit idea as Table 8, but with the unit of analysis collapsed to the agency level. This version is substantively closer to the institutional outcome because agency independence tier does not vary across administrations. A negative coefficient on mean distance implies that agencies further from the president tend to fall into more independent tiers.

4. Results

Figure 1. Conservative Trends in Agency Ideal Points

Figure 1 reproduces the central empirical finding of Acs (2025): several agencies shift in a conservative direction over administrations, with the largest movement concentrated in a small subset of agencies. Most agencies cluster near the ideological center across all four administrations, while a minority drift substantially. This pattern is consistent with the author’s argument that agency ideology is not static and responds to presidential incentives over time.

Table 1. Agencies with Largest Shifts in the Conservative Direction

Table 1 identifies the agencies with the largest minimum-to-maximum range in estimated ideal points across administrations. The agencies at the top of this list represent the subset where executive pressure or personnel turnover most strongly reshaped ideological orientation. The replication reproduces the agency-level ordering and substantive interpretation from Acs (2025)’s Figure 4.

Table 2. Ideological Distance and Regulatory Productivity

Table 2 replicates ACS Equation (3) across five outcome definitions. The coefficient on **distance** is negative across all columns, indicating that agencies whose ideology is further from the president produce fewer pages of final rules, fewer significant rules, and fewer regulatory-only rules. The negative binomial specification absorbs agency fixed effects and weights by the inverse variance of the ideal point estimates. The sign and magnitude patterns replicate the original Table 2 in Acs (2025).

The following subsections report results from the three methodological extensions. Each introduces a different outcome or estimation strategy to probe the scope of the ideological alignment mechanism identified by Acs (2025).

Table 6. ACS Distance and Fazekas Donation/Contract Outcomes

Table 6 extends ACS Equation (3) by replacing regulatory activity outcomes with donation and contracting outcomes from Fazekas et al. (2023). The central parameter in each column is the coefficient on **distance**. A negative coefficient indicates that greater ideological distance between the president and the agency is associated with lower donor-linked contracting intensity in that agency-administration cell. Stability of the sign across columns supports a more robust interpretation; sign reversals indicate sensitivity to outcome choice.

Figure 2. Distance Effect across Alternative Fazekas Outcomes

Figure 2 plots the **distance** coefficient from the extension models alongside 95% confidence intervals. This visual comparison identifies which outcome definitions produce the most precise estimates and whether the direction of the distance effect is consistent across donor contract share, donor contract count, log donation amount, and total contracts. A cluster of negative estimates with non-overlapping confidence intervals would provide the strongest evidence that ideological alignment shapes distributive outcomes in procurement.

Table 7. Logistic Regression of Donation Probability on Ideological Distance

Table 7 reports logistic regression estimates for the binary **donation_dummy** outcome. Column (1) includes only **distance** as a predictor; Column (2) adds an administration fixed effect. A negative coefficient on **distance** indicates that firms are less likely to donate when the agency’s ideology is far from the president, consistent with strategic targeting of politically aligned agencies. The divide-by-4 rule (Gelman et al. (2025)) converts the logit coefficient to an approximate upper bound on the marginal probability change per unit of ideological distance.

Figure 3. Predicted Donation Probability by Distance

Figure 3 translates the logistic model into predicted donation probabilities across the observed range of **distance**, separately for Bush and Obama administrations. A downward-sloping curve indicates that firms are less likely to donate as agencies become more ideologically distant from the president. The administration comparison shows whether this strategic-targeting pattern is stable across partisan contexts or specific to one presidency.

Figure 4. Overdispersion Diagnostic

Figure 4 compares the distribution of zero-rates across 200 simulated datasets from Poisson and negative binomial models fitted to the ACS regulatory productivity data, alongside the observed zero-rate (dashed line). The Poisson model generates replicated zero-rates systematically below the observed proportion, indicating that its constraint $\text{sd}(y) = \sqrt{\text{E}(y)}$ is too restrictive for these data. The negative binomial model generates replications that surround the observed zero-rate, validating the overdispersion assumption in Acs (2025)’s Equation (3).

Table 8. Ordered Logistic Regression of Agency Independence Tier

Table 8 reports the ordered logistic coefficient for **distance** on the four-tier agency independence scale, from most independent (indep. commission and IRC) to least independent (cabinet department). A negative $\hat{\beta}$ implies that agencies with greater ideological distance from the president tend to occupy more independent tiers, consistent with the institutional logic that structural independence reduces direct presidential alignment. A null result would indicate that ideological distance and formal independence are orthogonal properties of agencies in this sample.

Table 9. Agency-Level Ordered Logistic Regression of Independence Tier

Table 9 re-estimates Extension 3 after collapsing the data to one row per agency and replacing administration-specific distance with mean ideological distance across administrations. This specification is closer to the institutional outcome because agency independence tier is fixed at the agency level. If the coefficient remains negative and is similar in magnitude to Table 8, the cross-administration result reflects a stable cross-agency pattern. If it attenuates toward zero, the earlier result is more likely to reflect within-agency repetition from repeated agency-administration observations rather than a genuine structural relationship between ideological distance and independence tier.

5. Discussion

Implications

This extension moves the dependent variable from regulatory output to procurement outcomes, testing whether the same ideological alignment mechanism identified in Acs (2025) also shapes distributive politics in federal contracting from Fazekas et al. (2023). If ideological distance depresses both rule production and donor-linked contracting, this suggests that presidential-agency alignment is a general organizing principle of bureaucratic behavior rather than a phenomenon specific to rulemaking. Agencies that are ideologically close to the president are not only more productive in regulatory terms but also more responsive to political signals from contributing firms.

The logistic and ordered logistic extensions add two dimensions to this picture. The logistic result indicates whether firms actively sort their donations toward ideologically aligned agencies, while the ordered logit tests whether ideological distance is itself a function of an agency’s structural independence tier. Together, these three extensions separate the supply side (which agencies receive more regulatory output) from the demand side (which agencies attract more donation activity) of the political economy of bureaucratic alignment.

The agency-level follow-up to Extension 3 sharpens this interpretation by aligning the unit of analysis with the institutional outcome itself. If the relationship between ideological distance and agency independence persists after collapsing to one row per agency, the result is less likely to be an artifact of repeated agency-administration observations.

Method-Theory Link

The theory in Acs (2025) holds that ideological distance between the president and an agency reduces the president’s incentive to protect or prioritize that agency’s output, thereby depressing regulatory productivity. My extension translates this mechanism into a procurement context using Fazekas et al. (2023). If distance also reduces the probability that a firm donates to an ideologically distant agency, this indicates that the political economy of alignment operates at the firm level, not only at the level of agency output. The method follows directly from this theoretical chain: (1) logistic regression is the appropriate estimator when the outcome is a firm’s binary donation decision, (2) negative binomial is appropriate for count outcomes where overdispersion is present, and (3) ordered logistic is appropriate when the outcome is a categorical independence hierarchy with more than two ordered levels.

Limitations and Future Research

The principal limitation of this extension is the reliance on cross-dataset agency matching by name, which introduces measurement error when agency names differ across the Acs (2025) and Fazekas et al. (2023) data sources. To maximize panel coverage, I combine three matching strategies: strict matching against ACS agency identifiers, broad matching using Federal Register rulemaking records from 2004 to 2015, and fuzzy string matching for agencies unresolved by the first two approaches. Despite these efforts, the extension panel covers only the Bush and Obama administrations (2001-2016), as the Fazekas et al. (2023) dataset does not extend to the Trump period covered by Acs (2025). A more complete crosswalk using official agency code identifiers (RINs in the ACS data and USASpending codes in the Fazekas data) would further improve coverage and reduce matching error.

More substantively, the current extension treats **distance** as time-varying at the administration level but does not interact it with presidential party or other institutional factors that might affect agencies’ decision-making processes. Acs (2025) provides continuous ideal point estimates for 69 agencies across four administrations; a natural extension would interact these estimates with a partisan indicator or an agency ideology factor and examine whether ideological proximity amplifies donor-linked contracting (e.g., when the president and agency are aligned with the contributing party). I plan to explore this design in future work.

References

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